

Observation-Referenced Tissue Deformation Forecasting with Tool-Driven and Residual Dynamics

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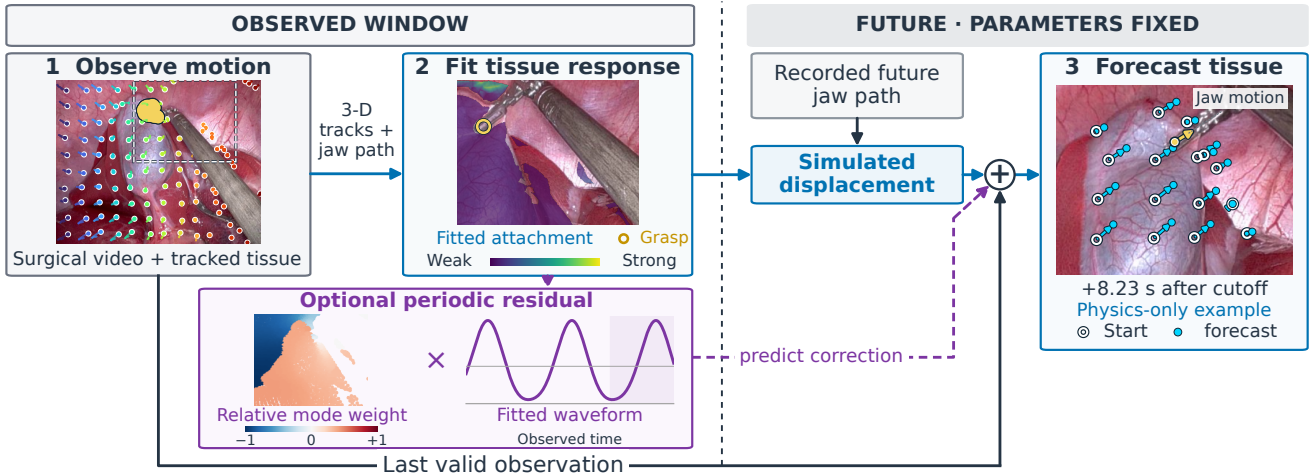


Fig. 1. Observe, fit, forecast on SM-P3. Tracked tissue and jaw motion define the input; the attachment map illustrates fitted coupling. Cyan arrows show physics displacement from white starting rings, yellow shows jaw motion, and the ring marks the grasp. The optional residual combines fixed spatial patterns with fitted oscillations. Parameters remain fixed after cutoff.

Abstract—Forecasting tissue deformation from instrument motion could support predictive planning in surgical robotics. We fit a scene-specific differentiable surface model to an observed tool-tissue interaction, then predict tissue motion under the recorded future jaw trajectory. An observation-referenced formulation adds simulated displacement to each point’s last valid observation, aligning the forecast with the measured surface while preserving the simulated state. The model fits spatially varying attachment, regional rigidity, region scales, and grasp coupling. An optional residual extension combines fixed graph-spectral patterns with fitted periodic dynamics. Across 12 manipulation clips, observation referencing reduces 3-D error relative to raw fitted forecasts with a median within-clip ratio of 0.598. Referenced physics improves on persistence in 8/12 clips, with a median ratio of 0.912 (95% interval 0.787–1.070). The periodic extension provides modest, clip-dependent gains in a three-clip study. These results examine observation-referenced forecasting as a predictive component for surgical scene models and digital twins.

I. INTRODUCTION

Predictive surgical scene models aim to anticipate how tissue will move when an instrument moves. Video-derived simulation methods increasingly reconstruct endoscopic interactions and their physical response [1]–[3]. For planning, the next question is whether a fitted model can forecast

tissue motion beyond the observations used to identify it. This requires separating alignment with the observed surface from prediction of its subsequent deformation.

Our approach combines an observed starting configuration with a model-predicted change (Fig. 1). We fit a differentiable, mesh-free simulator to the initial interaction [4], [5], then add its displacement to each point’s last valid observation. This cancels constant positional offsets while retaining the simulated response. An optional sequential residual predicts oscillatory fitting error, motivated by physiological motion [6].

We present the scene representation, gradient-based fitting, and referenced forecast construction, and evaluate their contributions on 12 clips from 10 surgeries. The future jaw path comes from the recording, so the task is tissue-response forecasting conditioned on known instrument motion. Later tissue coordinates are reserved for scoring; the reconstructed tracks themselves are produced offline. This scope lets us examine the forecasting component needed by predictive surgical twins.

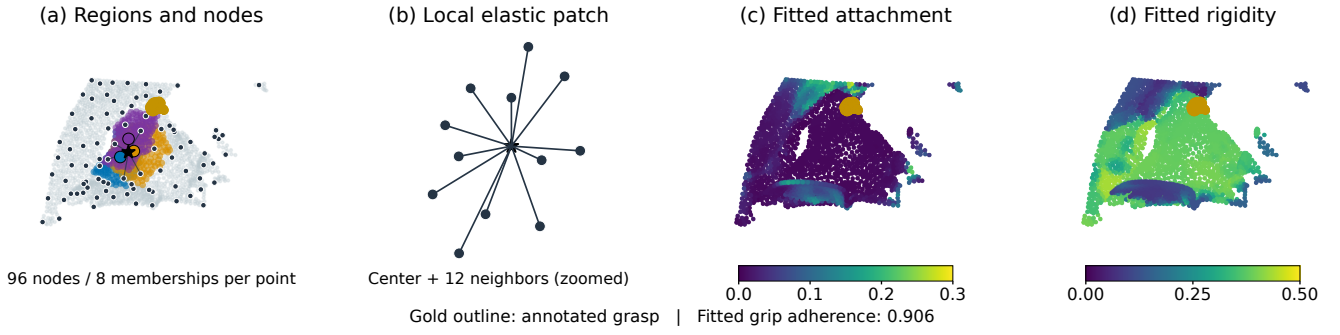


Fig. 2. Simulation structure and fitted coupling on SM-P3, in a camera-oriented 3-D view. (a) Scaffold nodes and example overlapping regions. (b) A local elastic patch. (c,d) Fitted per-point attachment and membership-weighted regional rigidity, starting from uniform values of 0.05 and 0.15. Gold marks the annotated grasp. These fields are dimensionless effective coupling weights fitted on the observed interval.

II. METHOD

We first construct a deformable surface from reconstructed tissue tracks, fit its tool-driven response, and then optionally fit residual dynamics with physics fixed. Given tissue observations $\mathbf{X}_{0:\tau}^{\text{obs}}$ and the instrument trajectory $\mathbf{U}_{0:T}$, we forecast frames $t > \tau$ as

$$\hat{\mathbf{X}}_{\tau+1:T} = \mathbf{X}^{\text{ref}} + \Delta \mathbf{X}^{\text{tool}} + \alpha \Delta \mathbf{R}^{\text{per}}. \quad (1)$$

The three terms are the last valid observed configuration, simulated tool-driven displacement, and an optional referenced periodic correction. All fitted parameters remain fixed during forecasting.

A. Scene construction and physical model

The tracked surface defines the model’s reference geometry and spatial couplings (Fig. 2). We use $N = 2000$ points, 12-neighbor elastic patches, and $M = 96$ farthest-point scaffold nodes. Each point belongs to its eight nearest nodes. Fixed neighborhood membership connects the surface, while fitted region scales determine how strongly each node influences its members. Local elasticity regulates short-range deformation, and overlapping shape matching [7] coordinates motion across regions.

Tool and tissue attachment constraints provide the external couplings. A manually annotated grasp set follows the tracked jaw translation through a shared adherence parameter g , with point offsets and jaw orientation fixed. Per-point weights a_i pull the surface toward its reference configuration. These are effective surface couplings inferred from displacement observations. Local elasticity uses the established stable Neo-Hookean formulation [8]–[10], with regularized local deformation gradients, within XPBD [11]. The simulation advances quasi-statically with 20 projection iterations per video frame, applying attachment, regional shape matching, elasticity, and grasp constraints.

B. Observation-referenced parameter fitting

Fitting adjusts the surface response to match observed tissue motion. The $2M + N + 1 = 2193$ parameters $\theta = \{\rho, \mathbf{r}, \mathbf{a}, g\}$ comprise regional rigidity, region scales, attachment, and

grasp adherence. Logistic parameterizations bound rigidity to $(0, 0.5)$, attachment to $(0, 1)$, and grasp adherence to $(0.05, 1)$; exponential parameterization makes region scales positive. Reference neighborhoods and grasp membership stay fixed.

The fitting objective compares observed motion with simulated changes from the same reference. Let $v_i(t)$ indicate observation validity and $t_i^* = \max\{t \leq \tau : v_i(t) = 1\}$ be point i ’s last valid observed index. With simulated positions $\mathbf{x}_i^{\text{sim}}(t; \theta)$, the referenced prediction is

$$\mathbf{q}_i(t; \theta) = \mathbf{x}_i^{\text{obs}}(t_i^*) + \mathbf{x}_i^{\text{sim}}(t; \theta) - \mathbf{x}_i^{\text{sim}}(t_i^*; \theta). \quad (2)$$

We minimize the masked squared trajectory error over the observed interval,

$$\mathcal{L}_{\text{fit}}(\theta) = \frac{\sum_{t=0}^{\tau} \sum_i v_i(t) \|\mathbf{q}_i(t; \theta) - \mathbf{x}_i^{\text{obs}}(t)\|_2^2}{\sum_{t=0}^{\tau} \sum_i v_i(t)}. \quad (3)$$

Only points with a valid reference contribute. Optimization uses a 30-epoch raw-trajectory warmup followed by 60 referenced epochs of Adam, with learning rate 0.03 and gradient clipping at 0.5. Gradients propagate through the full observed rollout, including both simulated terms in Eq. (2). eight-frame checkpoint segments limit memory use. The lowest-loss parameters define θ^* .

C. Forecasting from the last observation

The forecast retains the observed position and adds the simulated displacement. With θ^* frozen, the simulator continues under the recorded future jaw motion, and Eq. (2) defines its physics-only forecast for $t > \tau$. Referencing changes the output coordinates while preserving the simulated state. If $\mathbf{e}_i(t) = \mathbf{x}_i^{\text{sim}}(t) - \mathbf{x}_i^{\text{obs}}(t)$, the physics-only forecast error is $\mathbf{e}_i(t) - \mathbf{e}_i(t_i^*)$. Thus constant per-point offsets cancel, while displacement errors remain. Zero predicted displacement recovers persistence.

D. Optional periodic residual

The residual model predicts spatially coherent oscillations remaining after physics fitting. We use $K = 32$ fixed modes Φ_{ik} of the reference cloud’s symmetric 12-neighbor graph

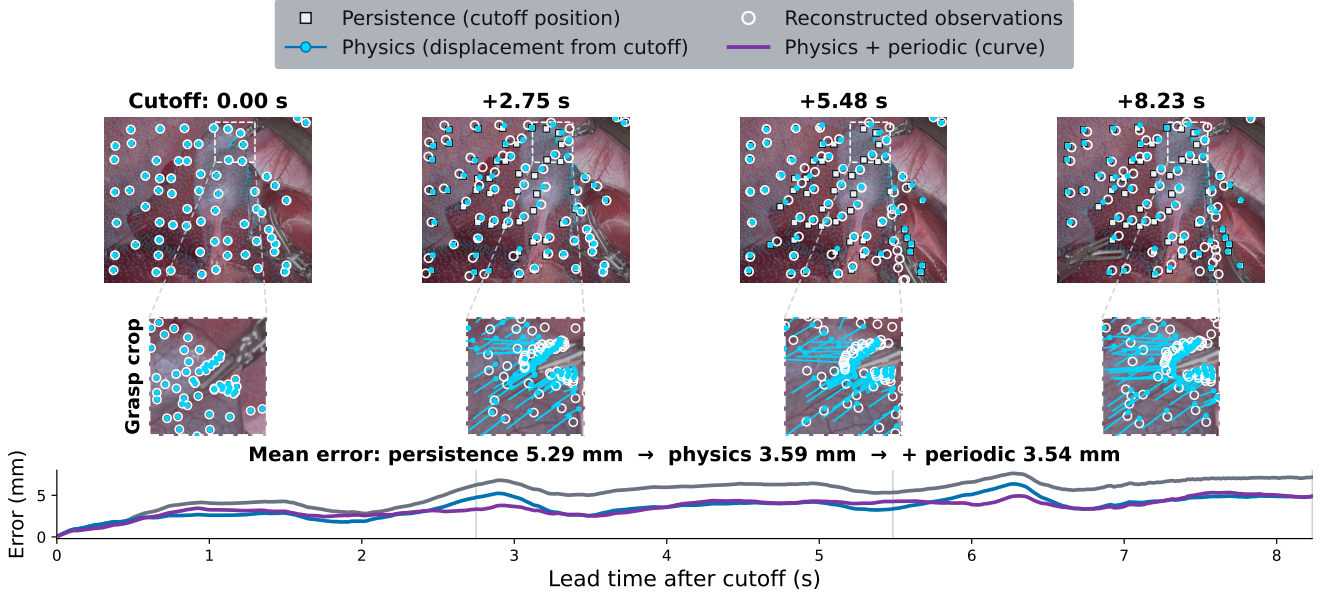


Fig. 3. SM-P3 at four uniformly spaced times. White rings show reconstructed observations, squares persistence, and cyan points referenced physics. Segments show displacement from cutoff to prediction. The second row enlarges the fixed grasp crop. Blue, gray, and purple curves report mean 3-D point error for physics, persistence, and the conditional periodic extension, with guides at the displayed times. The stated mean errors cover the full horizon.

Laplacian, following manifold harmonics [12]. Each mode has two temporal harmonics,

$$\mathbf{r}_i(t) = \sum_{k=1}^K \Phi_{ik} \sum_{h=1}^2 [\mathbf{a}_{kh} \cos(2\pi h\nu t) + \mathbf{b}_{kh} \sin(2\pi h\nu t)], \quad (4)$$

where the 3-D coefficients describe temporal amplitudes and ν is in cycles per frame. The forecast becomes $\hat{\mathbf{x}}_i(t) = \mathbf{q}_i(t; \theta^*) + \alpha[\mathbf{r}_i(t) - \mathbf{r}_i(t_i^*)]$. Referencing the correction preserves the same observed starting positions.

We fit the harmonic coefficients to the remaining referenced error $\mathbf{x}_i^{\text{obs}}(t) - \mathbf{q}_i(t; \theta^*)$ by masked least squares with a trace-scaled ridge penalty, at unit correction strength. A 70/30 temporal split within the observed prefix selects ridge strength and $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ by validation point error. Frequency is estimated from modal residual power on the fitting portion, using 512 candidates in $[1/200, 1/80]$ cycles per frame. The inherited frequency search zero-fills invalid residuals; coefficient fitting retains the masks. Frequency and coefficients are refitted on the complete observed prefix after selection.

This selection is conditional on physics already fitted to the complete observed prefix, including the residual-validation segment. A fully nested control also refits physics on the first 70% before selecting residual settings on the remaining 30%. Both variants use full-prefix physics and refitted residual coefficients for the final future forecast. Setting $\alpha = 0$ recovers physics alone.

III. EVALUATION PROTOCOL

The evaluation uses 12 development-selected grasp clips from 10 surgeries: nine monocular Open-H clips [13], two calibrated StereoMIS clips [14], and one in-house stereo record-

ing collected under UCSD HRPP #070449. AllTracker [15] tracks are lifted with FoundationStereo [16] or Video Depth Anything [17]. Instrument masks exclude tools from the tissue surface; manually marked instrument points define the jaw tracks. Reconstruction and visibility-based surface selection use the full recording. Only the two StereoMIS clips have calibrated metric scale; other 3-D distances are in nominal units. Cutoffs were fixed near the midpoint, with forecast horizons of 0.72–31.47 s; SM-P3 spans 8.23 s.

Comparisons share tracked points, observation cutoffs, and future validity masks. We cross default/fitted parameters with raw/referenced outputs to separate fitting from referencing. Persistence holds each last valid observation. Tool ridge fits a per-point linear map from jaw displacement to tissue displacement, with regularization selected within the observed prefix. All instrument-conditioned methods receive the same recorded future jaw path. The complete-corpus comparison uses physics alone; the residual extension is tested on three designated manipulation pilots.

The primary score is mean Euclidean distance over valid future point-frame observations, restricted to points observed by the cutoff. We report medians of within-clip error ratios and 95% percentile intervals from 10,000 surgery-cluster bootstrap draws (seed 8021), retaining all clips of each resampled surgery. The ten available clusters give a coarse estimate of uncertainty within this corpus. Ratios below one favor the numerator. A post-hoc image-space analysis projects the same forecasts and reconstructed observations into pixels, keeping cutoffs and masks fixed. We distinguish projecting the original 3-D ridge forecast from separately fitting ridge to 2-D targets.

Our referenced physics / baseline error

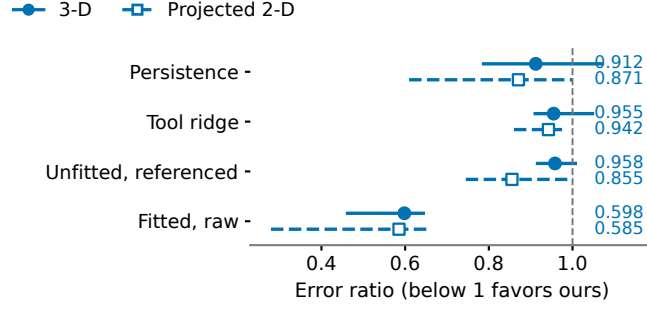


Fig. 4. Referenced fitted physics against four baselines across all 12 clips. Circles show 3-D error ratios; squares show camera-projected error ratios for the same forecasts, including the original 3-D-fitted ridge baseline. Bars are 95% surgery-clustered intervals. Values below one favor our forecast.

IV. RESULTS

A. Qualitative tissue forecasts

The calibrated SM-P3 example shows how predicted displacement updates the observed configuration (Fig. 3). Full-horizon mean 3-D error is 5.29 mm for persistence, 3.59 mm for referenced physics, and 3.54 mm with the conditional periodic extension. The fixed-view overlays retain true displacement scale; error curves show when the extension improves or worsens physics, complementing the aggregate results below.

B. Complete-corpus comparison

Observation referencing provides the largest measured 3-D error reduction. It improves 11/12 clips over raw fitted physics, with median ratio 0.598 (95% interval 0.462–0.643). When both default and fitted physics are referenced, the fitted-to-default ratio is 0.958 (0.916–1.007). After referencing, the additional fitting benefit is small and its interval includes parity.

Referenced physics improves on persistence in 8/12 clips, with median ratio 0.912 (0.787–1.070), and on tool ridge with ratio 0.955 (0.911–1.048). Both intervals include parity. In image space, it improves on persistence in 9/12 clips, with ratio 0.871 (0.609–1.000). Against ridge fitted directly in 2-D, the ratio is 0.948 (0.890–1.051); this is a separate sensitivity from the projected-ridge comparison in Fig. 4.

Clip-level outcomes depend on the scene and horizon. The calibrated clips SM-P2 and SM-P3 have ratios to persistence of 0.718 and 0.677, respectively, with SM-P2 limited to a 0.72 s forecast. On OH-747, physics has higher 3-D error than persistence (14.89 versus 11.91 nominal units), but slightly lower image-space error (24.35 versus 24.96 pixels). Depth accounts for 97% of the additional *mean squared* 3-D error relative to persistence. Independent geometry is needed to distinguish reconstruction error from simulation error.

C. Periodic extension

Conditional residual selection improves on physics in 2/3 pilots: SM-P3 by 1.4% and OH-747 by 4.9%, while SM-P2

worsens by 1.7%. The fully nested control improves 1/3 pilots. OH-747 remains worse than persistence with either extension. These results support a selectively applied residual correction, with limited evidence for a general improvement. The fitted manipulation frequencies lie at the search-band boundaries; identifying the oscillations as respiration would require an independent physiological signal.

V. DISCUSSION AND CONCLUSION

Observation-referenced forecasting combines measured tissue alignment with a scene-specific simulated response. Referencing provides the largest gain, while advantages over simple baselines and benefits from periodic correction remain clip-dependent. This supports further study of the method as a forecasting component for surgical digital twins.

Fitting and evaluation share offline reconstruction errors, the clips were development-selected, and horizons vary. The fixed-camera, translation-only grasp model omits rotation and slip. Untouched calibrated clips with independent geometry, followed by richer contact inputs and new instrument actions, are the next tests.

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